

Seongmoon Jeong

PH.D. CANDIDATE

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I am a Ph.D. candidate in the Department of Artificial Intelligence at Sungkyunkwan University, advised by **Prof. Jong Hwan Ko**. My research focuses on **image compression for machine perception**, spanning low-level optimization and high-level tasks such as classification, object detection, and segmentation. I develop **learned, task-aware representations** that jointly optimize compression efficiency and downstream vision performance. More recently, I have investigated high-level vision directly in the **RAW domain**, before the image signal processing (ISP) pipeline, to preserve information relevant to machine perception. I have authored or co-authored **eight peer-reviewed publications**, including papers at CVPR, AVSS, ICCAD, and in IEEE TCAS-I, and contributed **four ISO/IEC MPEG VCM proposals**. My long-term goal is to enable **communication-efficient intelligent systems** for autonomous systems and edge-based machine vision. **Available for full-time roles starting March 2027.**

Education

Ph.D. in Artificial Intelligence (Expected February 2027)

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Received an Outstanding Scholarship awarded to promising students in the Department of Artificial Intelligence.

B.S. in Electronic and Electrical Engineering

SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2013 - Feb. 2020

- GPA: 3.84/4.5 (Top 6.6%)

Research Experience

Graduate Researcher

IRIS LAB, SUNGKYUNKWAN UNIVERSITY (SKKU)

Suwon, Korea

Mar. 2020 - Present

- Machine-Oriented Image Compression:** Codec-transferable and test-time adaptation across downstream vision tasks [C4, C5]
- Task-Aware Preprocessing and JPEG Coding:** Adaptive downscaling, feature modulation, and region-based JPEG optimization [C1, C3, J1]
- RAW-Domain Machine Vision:** Foundation-model-guided RGB-to-RAW generation for object detection [C6]

Project Experience

Video Coding for Machines

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2023 - 2024

- Led research on codec-transferable adaptation for machine-oriented image compression [C5].

Adaptive and Efficient Preprocessing and Coding for Machine Vision Tasks and Input Images

LEAD STUDENT RESEARCHER (IRIS LAB., SKKU)

Media Research Division, ETRI

2022

- Led research on adaptive image downscaling for rate-accuracy-latency optimization [C3].

Multipurpose Visual Information Compression for Human and Machine Vision

PARTICIPATING RESEARCHER (BASIC RESEARCH LABORATORY, IRIS LAB., SKKU)

Ministry of Science and ICT (MSIT),
Korea

2025 - Present

- Contributing to unified visual compression methods supporting both human viewing and machine perception.

Selected Publications

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Hyon-Gon Choo, Jong Hwan Ko

IEEE AVSS

Sep. 2026

C4 Test-Time Fine-Tuning of Image Compression Models for Multi-Task Adaptability

Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

IEEE/CVF CVPR

Jun. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

IEEE AICAS

Apr. 2024

Standardization Activities

[VCM] Neural network based post-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71305

Geneva

Jan. 2025

[VCM] Neural network based pre-filter for VCM

ISO/IEC JTC1 / SC29 / WG4 M71304

Geneva

Jan. 2025

[VCM] Learned joint filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69990

Kemer

Nov. 2024

[VCM] Learned pre-filter network for VCM

ISO/IEC JTC1 / SC29 / WG4 M69085

Sapporo

Jan. 2024

Honors & Awards

2024 **3rd Place**, DAC System Design Contest - GPU Track

San Francisco, USA

2020 **2nd Place**, Artificial Intelligence Grand Challenge, Ministry of Science and ICT

Korea

2019 **Honorable Mention**, Big Data Center Open Innovation Challenge

Seongnam, Korea

2019 **Outstanding Work Award**, Capstone Design Project, SKKU

Suwon, Korea

2018 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

2017 **Dean's List Award**, Fall semester, SKKU

Suwon, Korea

Full Publications

INTERNATIONAL CONFERENCE

C6 Foundation Model-Guided RGB-to-RAW Generation with Spectral Supervision for RAW-Domain Detection

IEEE AVSS

Seongmoon Jeong, Yulhwa Kim, Jong Hwan Ko

Sep. 2026

C5 UICAM: A Codec-Transferable Adapter for Machine-Oriented Image Compression

IEEE AVSS

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Sep. 2026

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Un Ki Park, Seongmoon Jeong, Youngchan Jang, Gyeongmoon Park, Jong Hwan Ko

Jun. 2025

C3 Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression

IEEE AICAS

Hangyul Choi, Seongmoon Jeong, Sangwoon Kwak, Soon-heung Jung, Jong Hwan Ko

Apr. 2024

C2 Kernel Shape Control for Row-Efficient Convolution on Processing-In-Memory Arrays

IEEE/ACM ICCAD

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

Oct. 2023

C1 Rate-Controllable and Target-Dependent JPEG-Based Image Compression Using Feature Modulation

IEEE ICME Workshops

Seongmoon Jeong, Kang Eun Jeon, Jong Hwan Ko

Jul. 2023

INTERNATIONAL JOURNAL

J2 KERNTROL: Kernel Shape Control Toward Ultimate Memory Utilization for In-Memory Convolutional Weight Mapping

IEEE TCAS-I

Johnny Rhe, Kang Eun Jeon, Joo Chan Lee, Seongmoon Jeong, Jong Hwan Ko

Dec. 2024

J1 An Overhead-Free Region-Based JPEG Framework for Task-Driven Image Compression

Pattern Recognition Letters

Seonghye Jeong, Seongmoon Jeong, Simon S. Woo, Jong Hwan Ko

Jan. 2023

Research Engineering Experience

ML Systems & Research Infrastructure

IRIS LAB., SUNGKYUNKWAN UNIVERSITY

- **Distributed evaluation:** Built a Python-based multi-node inference system for large-scale CVPR experiment evaluation, enabling parallel execution and faster research iteration.
- **Input pipeline optimization:** Profiled data-loading and augmentation bottlenecks and optimized pipelines using TensorFlow Datasets and Grain, reducing input stalls during neural network experiments.
- **Reproducible experimentation:** Standardized environments and automated workflows using Docker, Conda/Mamba, uv, and Pixi, improving consistency across research machines.

Student Mentoring

Hangyul Choi

M.S. STUDENT AT SUNGKYUNKWAN UNIVERSITY, NOW AT MX BUSINESS, SAMSUNG ELECTRONICS

2023 - 2024

- Adaptive Image Downscaling for Rate-Accuracy-Latency Optimization of Task-Target Image Compression [C3]

Chanung Park

GRADUATE STUDENT AT SUNGKYUNKWAN UNIVERSITY

2024 - Present

- Mentored and collaborated on research in Video Coding for Machines (VCM) and multipurpose visual information compression through the BRL project.

Skills

Programming Languages	Python, Shell scripting, C/C++
ML Frameworks	PyTorch, TensorFlow, JAX, TensorRT
Development Frameworks	Docker, Conda, uv

Reference

Jong Hwan Ko

ASSOCIATE PROFESSOR

Ph.D. advisor

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